Eco Automation System Using IoT and OpenCV

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Abstract

This project aims to enhance energy efficiency in living spaces by optimizing the use of lights, and fans. By employing an IoT-based approach with real-time video monitoring and automation, the system intelligently controls energy consumption based on occupancy detection using OpenCV.

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1 Introduction

Energy efficiency in modern living spaces is crucial for reducing utility costs and environmental impact. Traditional systems often leave lights and appliances on when not needed. This project addresses this issue through intelligent automation based on occupancy detection.

1.1 Objectives

- Reduce energy consumption by automating lighting and fan controls.
- Provide a manual control interface via a web page.

2 Ideology

The core idea behind the Eco Automation System is to optimize energy consumption in living spaces by automating the control of lights, and fans. The system uses a combination of IoT devices and real-time video monitoring to detect the presence of people in a room, and automatically adjusts the state of appliances accordingly. The system also includes manual controls through a web interface for cases where the user needs direct control over the devices.

2.1 Motivation

With rising concerns about energy efficiency and sustainable living, this system aims to significantly reduce unnecessary energy consumption by only activating appliances when they are needed. The integration of OpenCV allows the system to accurately detect human presence in the space, ensuring lights and fans are only turned on when necessary.

3 Requirements

3.1 Hardware

- CCTV camera (for capturing live footage)
- ESP32 microcontroller
- 5V 1-channel relay module
- Bulb or other appliance to control
- Wi-Fi connection for ESP32
- Power supply (5V charger for ESP32)

3.2 Software

- Python (for programming logic)
- OpenCV (for human detection)
- Real-Time Streaming Protocol (RTSP)
- TCP/IP protocol (for communication between ESP32 and web interface)
- Web framework (for manual control interface)

4 Pros and Cons

4.1 Advantages

- **Energy Efficiency**: The system reduces unnecessary energy consumption by automatically controlling appliances based on occupancy.
- **Automation**: Real-time automation ensures seamless control without manual intervention.
- **Cost-Effective**: The use of affordable IoT components like ESP32 and relay modules makes the system accessible and cost-effective.

4.2 Disadvantages

- **Dependence on Wi-Fi**: The system's reliance on Wi-Fi can be a limitation in areas with poor connectivity.

5 Demo

For a demonstration of the Eco Automation System, I set up a simple system at home. The components used in this demo include a CCTV camera, a bulb, ESP32, and a 5V relay module. You can view the demonstration video at the following URL: `https://drive.google.com/file/d/1GHMg2Rck6GrDwDvuk_OXYMgy32GJM-KX/view?usp=sharing`

5.1 Setup

1. The CCTV camera was set up to monitor a corridor, and the RTSP URL was obtained for capturing live footage.
2. The ESP32 was connected to the relay module and configured to control the light bulb.
3. A Wi-Fi connection was established to enable communication with the ESP32 using TCP protocol.
4. The Python code was written to detect human presence using OpenCV. If a person is detected within a specific boundary (a blue box on the live footage), the light turns on.

Figure 1: Demo: nothing is Detected

Figure 2: Demo: Human Detection - Light ON

5. After 25 seconds of no detected presence, the system automatically turns off the light.

6 Budget

Item	**Description**	**Cost (INR)**
CCTV Camera	Basic IP camera with RTSP support (CP plus 360)	4700/-
ESP32 Microcontroller	ESP32 module for IoT connectivity	560/-
Relay Module	5V 2-channel relay module	200/-
Light Bulb	Standard LED bulb 10W	500/-
Power Supply	5 volt adaptor	150/-
Miscellaneous Wiring	Cable, current wire	150/-
Total		**5850/-**

Table 1: Budget for Eco Automation System at home

7 Demo Output

The demo system successfully automates the light control based on human presence. When a person enters the room, the system detects them and turns the light on. After the person leaves, the light turns off after a 25-second delay. The system's performance during the demo showcases the efficiency of using IoT and OpenCV for energy management in living spaces.